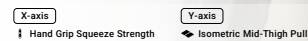
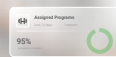
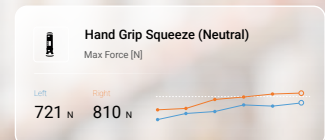


Quiet Stand

Avg Total Excursion - Bilateral Avg

105 mm

↑ 11.1%



Special Thanks

We would like to extend a special acknowledgment to [Dale Turner](#) and the team at [EMP Onsite](#) for their contribution to this guide. Their experience delivering large-scale, data-driven workplace wellness programs has provided valuable practical insight into how objective assessment can be implemented in real-world occupational settings. Their willingness to share both processes and outcomes has helped strengthen the applied focus of this resource.



Dale Turner

Chief Executive Officer

EMP Onsite

[in](#) [ig](#) [@emponsite](#)

“ The ability to act quickly on risk improves outcomes for both individuals and groups in workplace wellness settings. For far too long, insights have been sporadic and uncovered only when disaster hits.

Creating an environment where data builds risk profiles, provides meaningful insights and supports timely intervention drives value for all parties.

This is not the future; it is the here and now, and it will only get better. ”

Some Words from the Experts



Dr. Pierre Zook

Owner and Physical Therapist

Trek Physical Therapy

[in](#) [@trekphysicaltherapy](#)

“ VALD testing plays a key role in our corporate wellness programs by delivering fast, objective insights into an individual’s physical health. Strength and power are essential qualities for supporting longevity and maintaining independence with age, and this technology allows us to measure them with a high level of precision. Much like traditional lab testing, individuals receive clear, immediate results, making their health status tangible, trackable and actionable.

Because this type of strength and power assessment is not typically included in primary care visits, our programs offer a unique opportunity for both education and meaningful health insight.

We partner with forward-thinking organizations that invest in their employees’ health, giving them clear insight into where they are today and what is needed to improve, so they can perform at a high level both at work and throughout their lives. ”



Trey Willis

Owner

Semper Fi Mobility and Fitness

[@mfsemperfi](#)

“ HumanTrak and the VALD ecosystem took my sessions to another level. Technology has helped me deliver meaningful insights to my clients and ultimately get real results. Clients get to see their progress immediately, helping them buy in and get more out of their care. ”

Contents

Introduction	5
What You Will Gain from This Guide	7
Understanding Workplace Wellness	8
The Underlying Problem	13
Early Intervention and Workforce Profiling	18
Workplace Wellness Service Models	20
Long-Term Value in Workplace Wellness Programs	23
Group Testing and Reporting	24
Testing by Population	28
Case Study	31
What Next?	33
Appendix	34

In this document, we will **cover a wide range of topics and answer some common questions**, including:

Click any heading to jump to section

Wellness by the Numbers

Revenue Opportunities for Practitioners

Data Accessibility and Analysis

On-Site Testing Events

**Using Norms to
Support Interpretation**

Extending Value Beyond the Testing Day

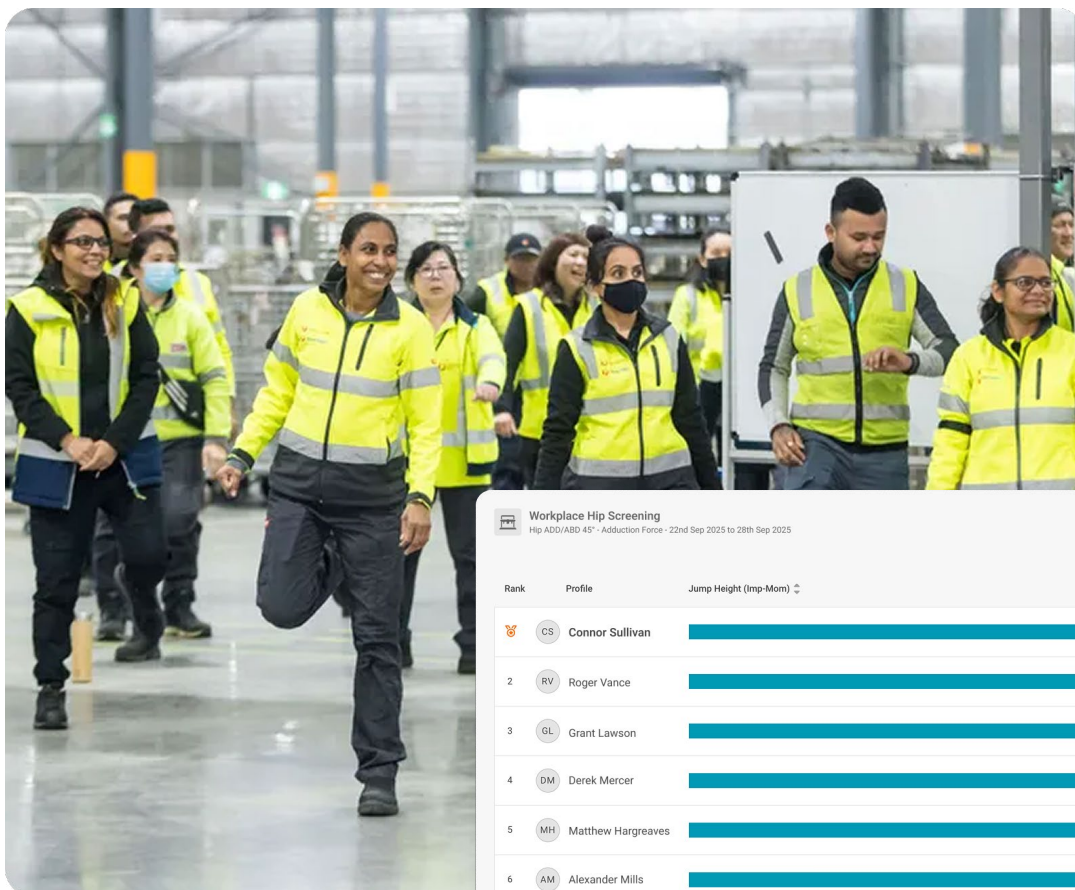
Introduction

Musculoskeletal injuries remain among the leading contributors to lost work time and healthcare expenditures across a wide range of industries. Despite this, many labor-specific organizations continue to rely on reactive care models, with limited use of objective methods to assess employee physical capacity or identify modifiable risk factors.

Practitioners working in rehabilitation and performance settings are already equipped to address these challenges. The same principles used to assess strength, movement and neuromuscular capacity in sport and clinical populations can be applied to occupational environments to better understand an individual's ability to tolerate the physical demands of their role.

The same principles used to assess strength, movement and neuromuscular capacity in sport and clinical populations **can be applied to occupational environments...**

Workplace wellness programs provide a practical pathway to extend these services beyond traditional settings. By implementing structured assessment and targeted intervention strategies, practitioners can support organizations in improving employee health, reducing injury risk and maintaining productivity, while simultaneously expanding their own service offerings.



This opportunity spans a wide range of industries, including manufacturing, construction, commercial driving, logistics, warehousing, healthcare, police and fire services, as well as corporate office environments. Each setting presents unique physical demands, but all benefit from the ability to objectively assess and develop workforce capacity.

Workplace wellness represents a **scalable opportunity for practitioner-owned businesses...**

Workplace wellness represents a scalable opportunity for practitioner-owned businesses, in which efficient testing, clear communication and practical interventions can deliver meaningful value to both employees and organizations.



What You Will Gain from This Guide

This guide outlines a practical framework for practitioners looking to expand into workplace wellness. It focuses on identifying service opportunities, structuring scalable assessment models and creating clear pathways to drive value for employers and employees.

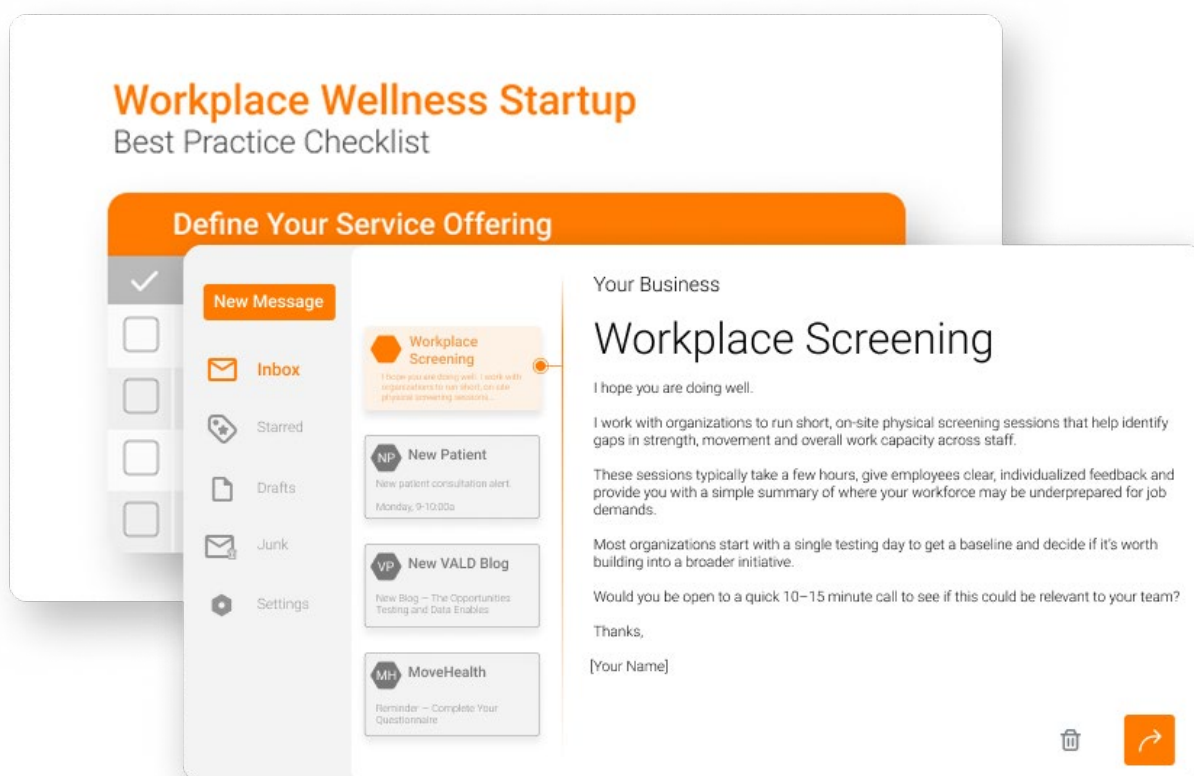
[This guide will identify] service opportunities, [structure] scalable assessment models and **[create] clear pathways to drive value for employers and employees.**

Readers will leave with the following:

- Understanding of how workplace wellness services are delivered
- Framework for selecting efficient and meaningful assessments
- Insight into how to communicate results and drive intervention
- Practical tools to support outreach, implementation and business development

The goal is to provide a clear and actionable starting point for integrating workplace wellness into existing practice.

See Appendix for Key Resources



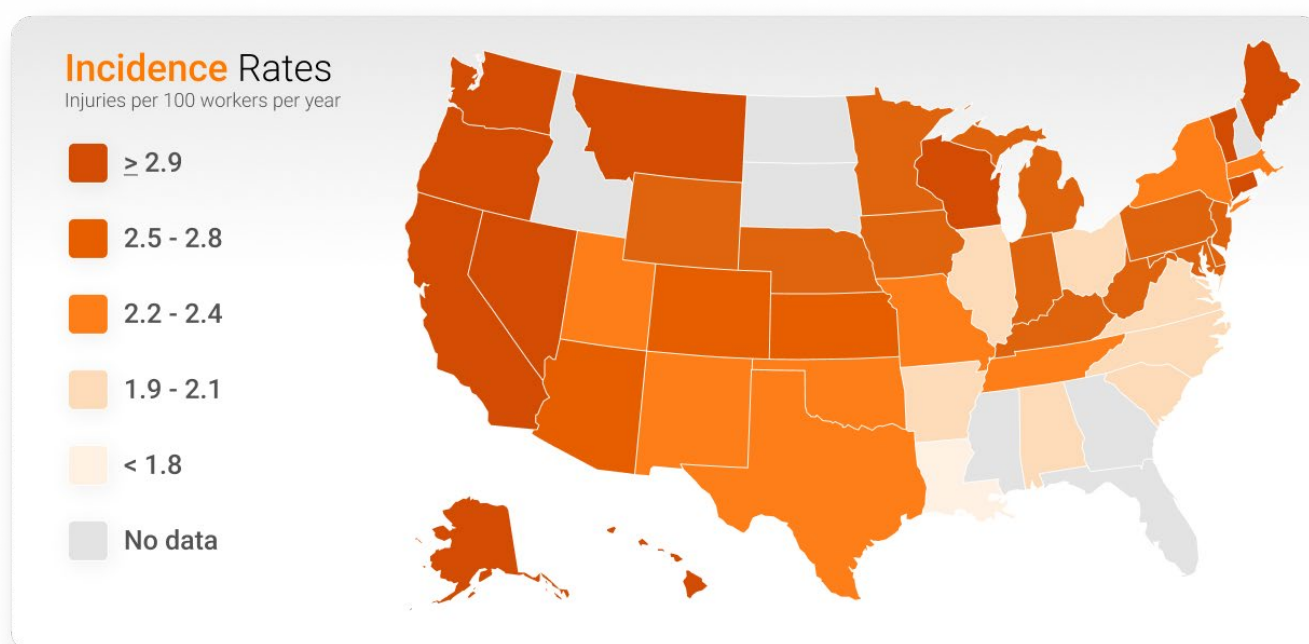
Understanding **Workplace Wellness**

Workplace wellness is best understood as a response to the measurable physical, operational and financial burden created by workplace injury.

Wellness by the **Numbers**

Workplace injury remains a significant and measurable burden in the U.S. and around the world. In the 2024 period, [U.S. private industry employers](#) reported 2,488,400 nonfatal workplace injuries and illnesses, including 1,834,600 cases that resulted in days away from work, with a median of 8 days lost per case.

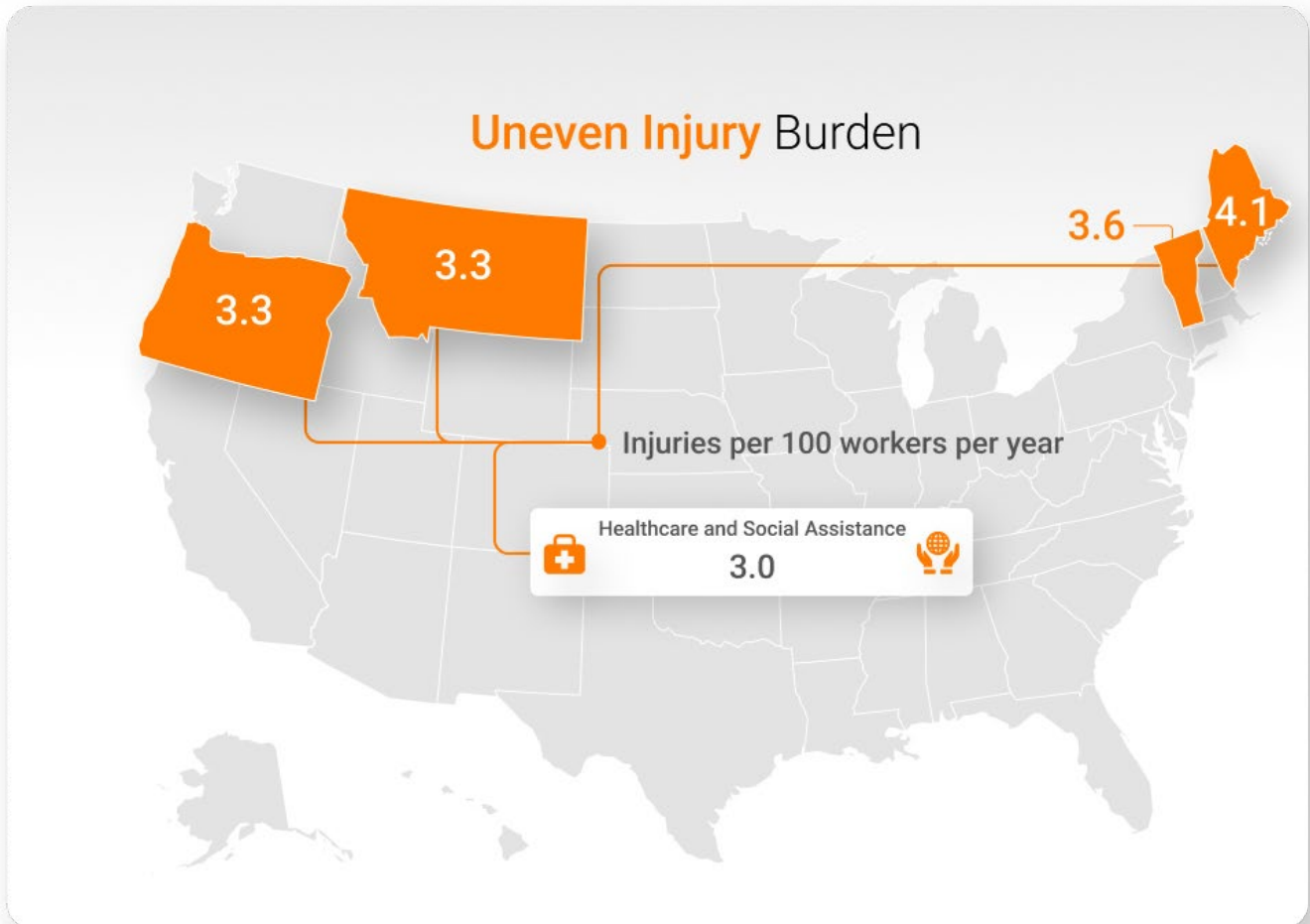
...U.S. private industry employers reported **2,488,400 nonfatal workplace injuries** and illnesses...



When translated into workforce impact, that equates to approximately 14.7 million lost workdays per year from days-away-from-work cases alone, or roughly 117.4 million work hours lost annually, assuming standard 8-hour shifts. Importantly, this estimate is conservative, as it does not include restricted duty, reduced productivity while still at work or added operational burden on surrounding staff and systems.

...restricted duty, reduced productivity while still at work or added operational burden on surrounding staff and systems [also affect workplace wellness].

This burden is not evenly distributed across industries or states. In 2024, the total recordable case rate was 3.0 per 100 full-time employees in healthcare and social assistance. On a [state-by-state basis](#), Maine (4.1), Vermont (3.6), Oregon (3.3) and Montana (3.3) had some of the highest private industry injury rates in the U.S. This reinforces that physically demanding sectors and operational environments continue to carry a disproportionate injury burden.



The cost of workplace injury extends well beyond missed workdays. According to the [National Safety Council \(NSC\)](#), U.S. work injuries cost employers an estimated \$181.4 billion in 2024.

...U.S. work injuries cost employers an estimated **\$181.4 billion in 2024.**

Workplace wellness becomes relevant because many workplace injuries are influenced by modifiable physical factors such as insufficient strength capacity, movement limitations, reduced balance and poor tolerance for repeated physical demands. However, these qualities are rarely measured proactively in occupational settings.

Using the NSC's data, a 10% reduction in annual workplace injuries would equate to nearly 250,000 fewer reported cases, preserve roughly 1.4 million workdays or 11.2 million work hours and represent an estimated \$18.1 billion in avoided annual employer costs. Workplace wellness exists to address this gap between occupational demand and physical capacity through objective assessment, early identification and targeted intervention.

 **2,488,400** 

Nonfatal workplace injuries and illnesses in U.S. private industry in 2024.



That leads to...



14.7 million
Lost workdays per year



117.4 million
Lost work hours per year

The **true cost** of workplace injury...



\$181.4 billion



Annual cost of U.S. work injuries

If annual workplace injuries were **reduced by just 10%**

250,000

Fewer cases

1.4 million

Workdays preserved

11.2 million

Hours saved

\$18.1 billion

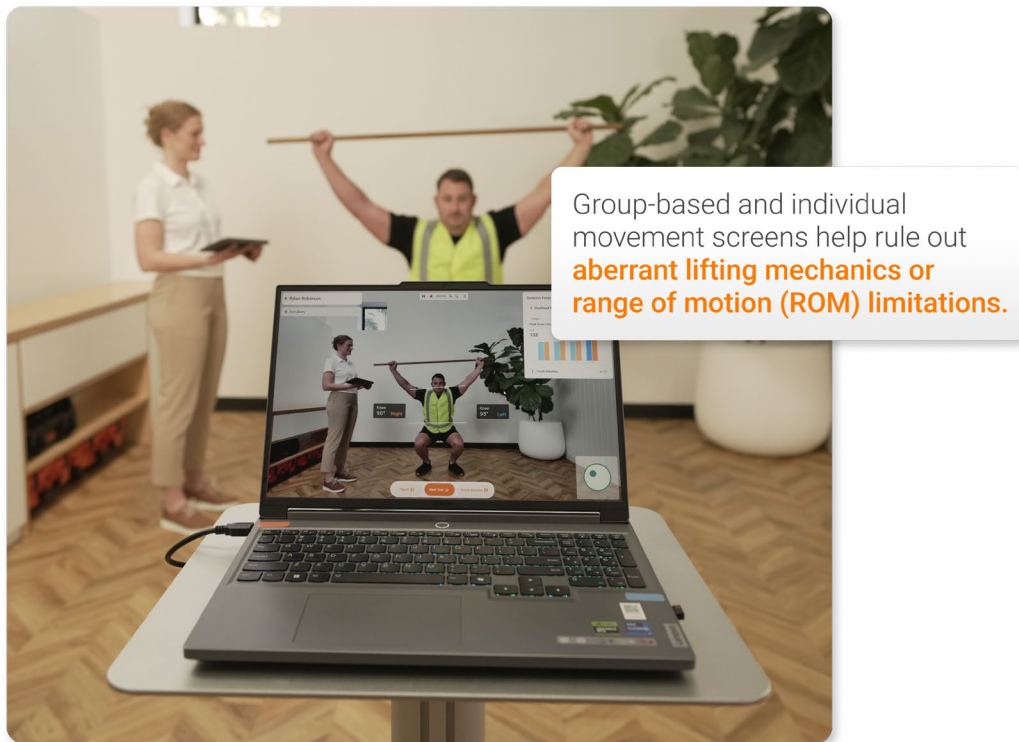
Employer costs saved

Revenue Opportunities for Practitioners

The financial burden of workplace injury reflects a substantial opportunity for practitioners. With employers **absorbing billions annually** in injury-related costs, there is a clear demand for services that support prevention and early intervention through objective assessment and targeted programming.

Workplace wellness enables practitioners to deliver services more efficiently than in traditional one-on-one care. Group-based testing, combined with clear reporting and structured follow-up, creates an efficient and scalable model that can extend into recurring partnerships through retesting, exercise programming and ongoing support.

Group-based testing, combined with clear reporting and structured follow-up, **creates an efficient and scalable model...**



Simple examples may involve:

- One on-site testing session for 40-60 employees
- Quarterly retesting for the same organization
- Immediate reporting and exercise programming delivered via [MoveHealth](#)
- Optional individual assessments for employees requiring further evaluation

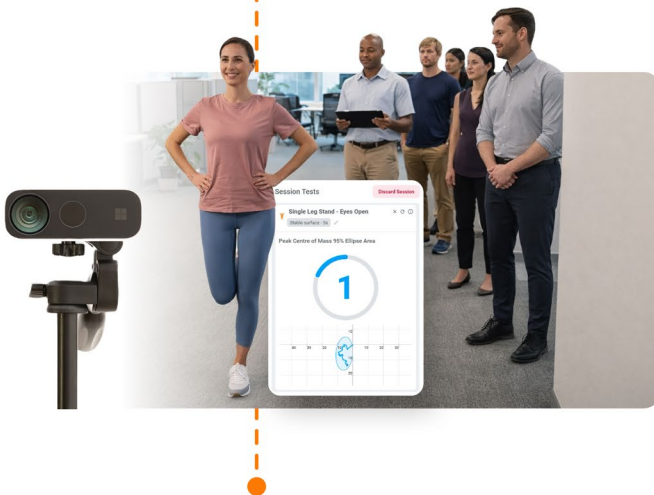
This allows workplace wellness to generate both immediate revenue and recurring service opportunities, while creating a pathway toward longer-term employer relationships and non-insurance-based business growth.

Revenue Calculator

On-Site Testing Day

50 employees
× \$100 per employee

\$5,000 revenue



Recurring quarterly assessments
for the entire company

\$5,000 × 4 quarters

\$20,000 annual revenue

Individual assessments and
treatment plans for employees
who had **increased risk of injury.**

8/50 employees flagged
for assessment



4 follow-up treatments
per assessment



\$75 per visit



8 assessments

32 follow-up appointments

\$75 per visit

\$3,000 in visit revenue

The Underlying Problem

Many workplace injuries are influenced by modifiable physical factors such as insufficient strength capacity, movement limitations, reduced fatigue tolerance and asymmetries. These qualities can be objectively assessed using technologies like force plates (e.g., [ForceDecks](#)), dynamometers (e.g., [DynaMo](#) and [ForceFrame](#)) and motion capture tools (e.g., [HumanTrak](#)) to better understand an individual's physical capacity relative to the demands of their role.



Despite this, most workplaces still rely on reactive, post-injury care, self-reported symptoms or generalized wellness initiatives that lack objective measurement. This creates a clear mismatch between job demands and employee capacity, as well as a gap in quality care and services that employees in demanding jobs require and deserve.

[There is] a clear **mismatch between job demands** and employee capacity...

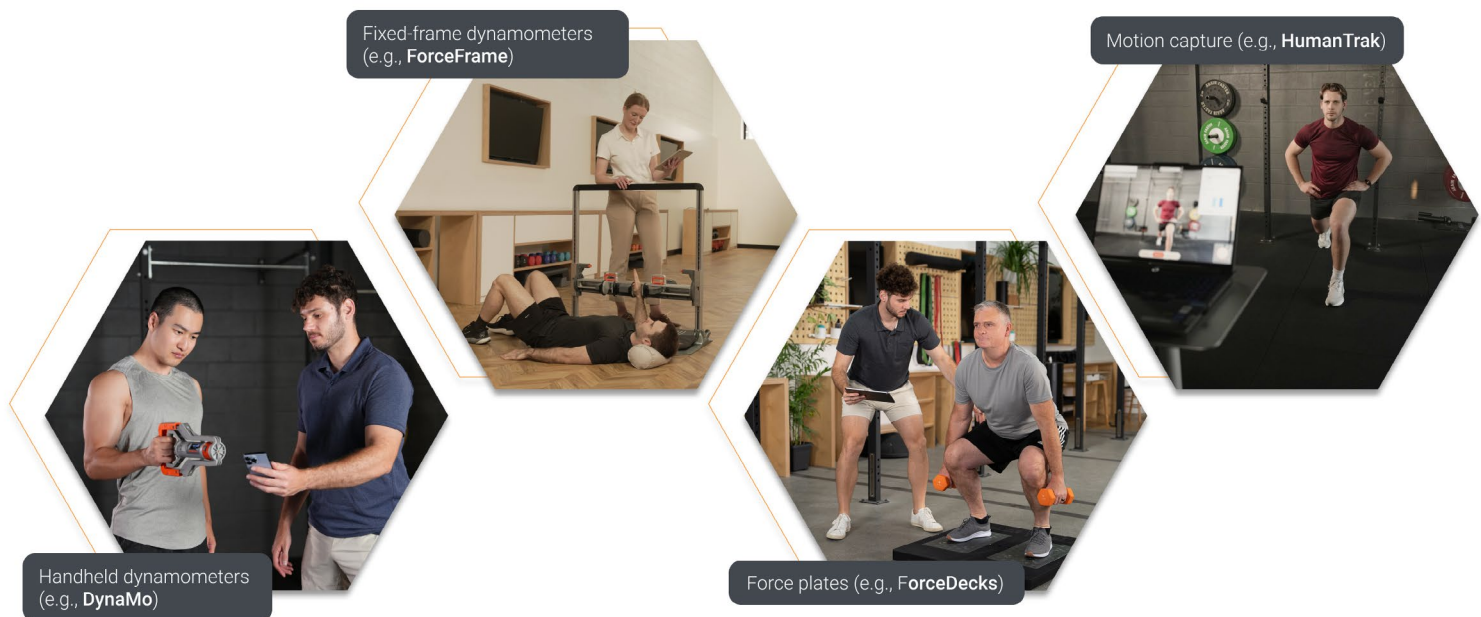
This issue often occurs due to a lack of awareness, planning and perceived access to these tools within organizations performing large-scale, labor-intensive work.

Workplace Wellness

Assessment Technologies

A range of objective assessment technologies can be used to quantify physical capacity in workplace settings. Selection should prioritize tools that are efficient, reliable and scalable across large groups while providing outputs that can guide decision-making. Common options include the following:

- **Handheld dynamometers** (e.g., [DynaMo](#)) provide a portable option for quick, isolated strength assessment across large groups.
- **Fixed-frame dynamometers** (e.g., [ForceFrame](#) and [NordBord](#)) enable standardized setup for more controlled and repeatable muscle group testing.
- **Force plates** (e.g., [ForceDecks](#)) assess power, balance and movement strategies across a range of assessments.
- **Markerless motion capture** (e.g., [HumanTrak](#)) measures joint angles and movement patterns during functional tasks.



Selection should **prioritize tools that are efficient, reliable and scalable** across large groups while providing outputs that can guide decision-making.

In practice, these technologies are often combined to create a clearer picture of strength, movement and neuromuscular capacity in relation to workplace demands

Infrastructure and Planning

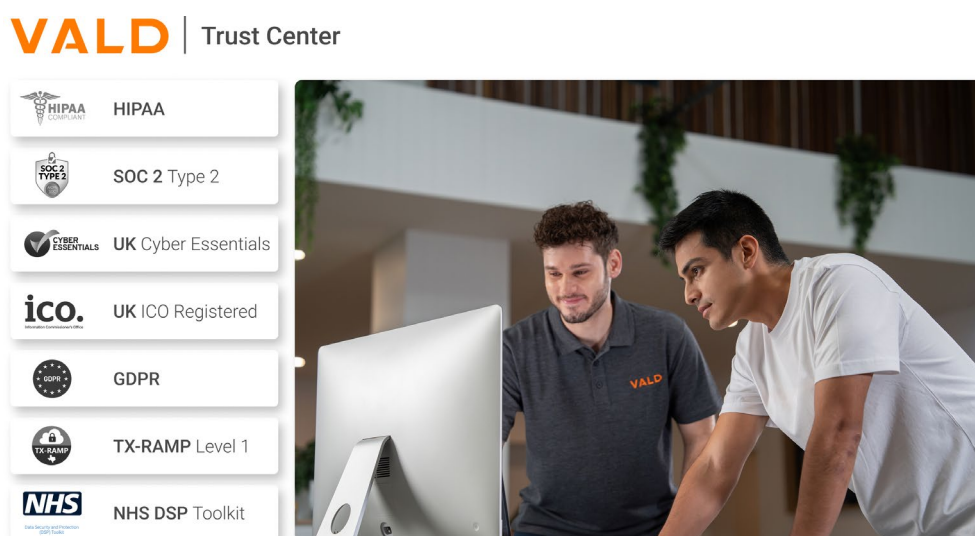
Effective workplace wellness programs require simple infrastructure that allows objective assessment to be delivered consistently across employees and over time.

Practitioners can create a basic infrastructure by selecting relevant tests, organizing efficient group-based workflows and ensuring that data is captured and securely stored in a centralized system such as [VALD Hub](#).

Large-scale organizations are likely to require greater security measures, including stricter data governance, controlled access permissions and compliance with recognized data protection standards. Platforms such as VALD Hub are designed to support these requirements, aligning with established standards such as the [Health Insurance Portability and Accountability Act \(HIPAA\)](#), [Service Organization Control 2 Type 2 \(SOC 2 Type 2\)](#) and the [General Data Protection Regulation \(GDPR\)](#).

Large-scale organizations are likely to require greater security measures... [Therefore], **VALD Hub [is] designed to...[align] with established standards** such as HIPAA, SOC 2 Type 2 and GDPR.

This security provides practitioners and organizations with confidence that assessment data is managed and handled in accordance with widely accepted regulatory standards. For more information about data security and compliance, please visit trust.vald.com.



Planning should focus on making assessment practical and repeatable within the demands of the workplace. This includes defining how often testing will occur, how employees will be grouped and how results will be used to guide follow-up. When these elements are in place, objective testing can become a consistent process that helps identify reduced capacity, guide intervention and monitor change across the workforce.

For more details on specific plans, workflows and testing operations, see the [Appendix](#).

Data Accessibility and Analysis

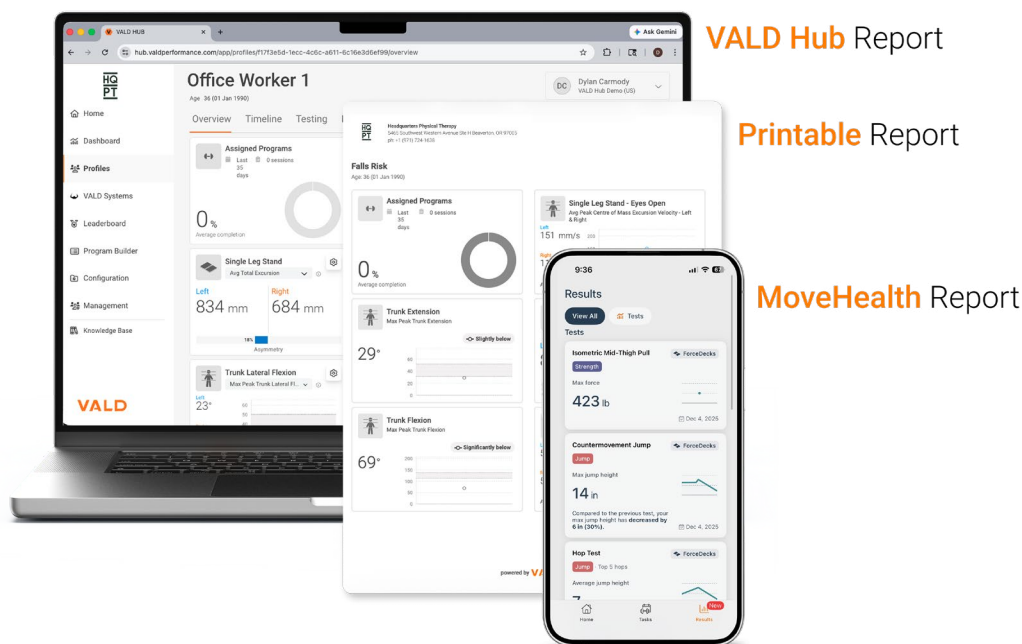
Access to clear, centralized data is critical for delivering and scaling workplace wellness services. Platforms such as **VALD Hub** allow practitioners to consolidate results from multiple systems into a single interface, making it easier to review individual performance, track changes over time and compare outcomes across teams or departments.

...VALD Hub [allows] practitioners to...**track changes over time and compare outcomes** across teams or departments.



To read this and all other VALD articles, visit valdhealth.com/news.

Features such as **timeline views**, **group dashboards** and **customizable reporting** help practitioners build a structured understanding of workforce capacity at both a group and individual level.



Practitioners can review sessions, identify trends, generate reports and communicate findings without manually compiling data across systems, improving efficiency and overall care value.

Group-level visualizations, such as [Group and Quadrant Reports](#) in [VALD Hub](#), support comparisons across roles or sites, while longitudinal tracking guides retesting and intervention planning. This ensures testing data is consistently used to inform decisions rather than remaining isolated within individual sessions.

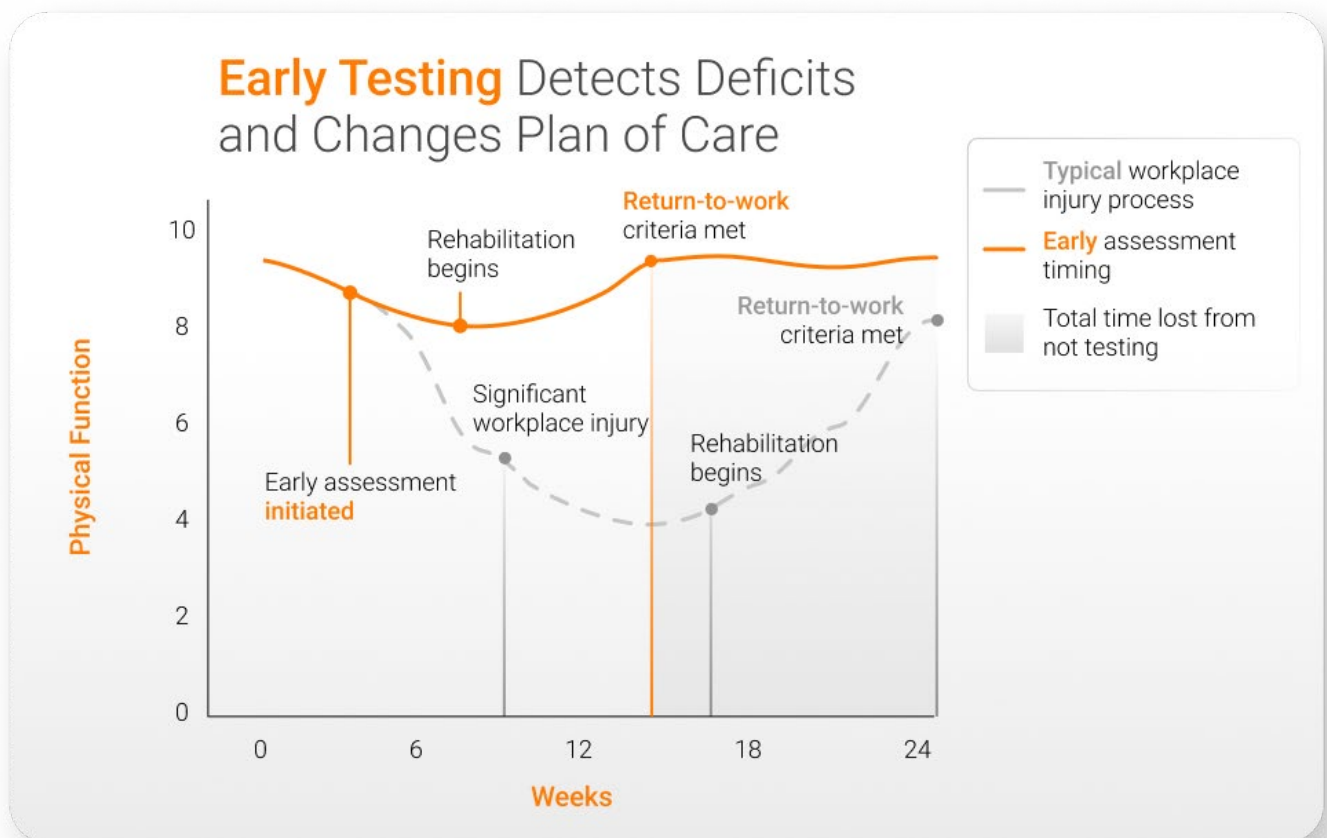
...Group and Quadrant Reports in VALD Hub, support comparisons across roles or sites, while **longitudinal tracking guides retesting and intervention planning.**

For most practitioners, data can be managed within existing practice management systems (PMS) or electronic medical records (EMRs), allowing assessment results to be stored, reviewed and integrated into standard workflows at both the individual and group level. For larger organizations or those with more advanced data capabilities, VALD's [application programming interface \(API\)](#) enables the extraction of larger datasets. These can be visualized and analyzed using tools such as Power BI or Domo to report trends across sites, departments and roles and to identify where support is required.



Early Intervention and Workforce Profiling

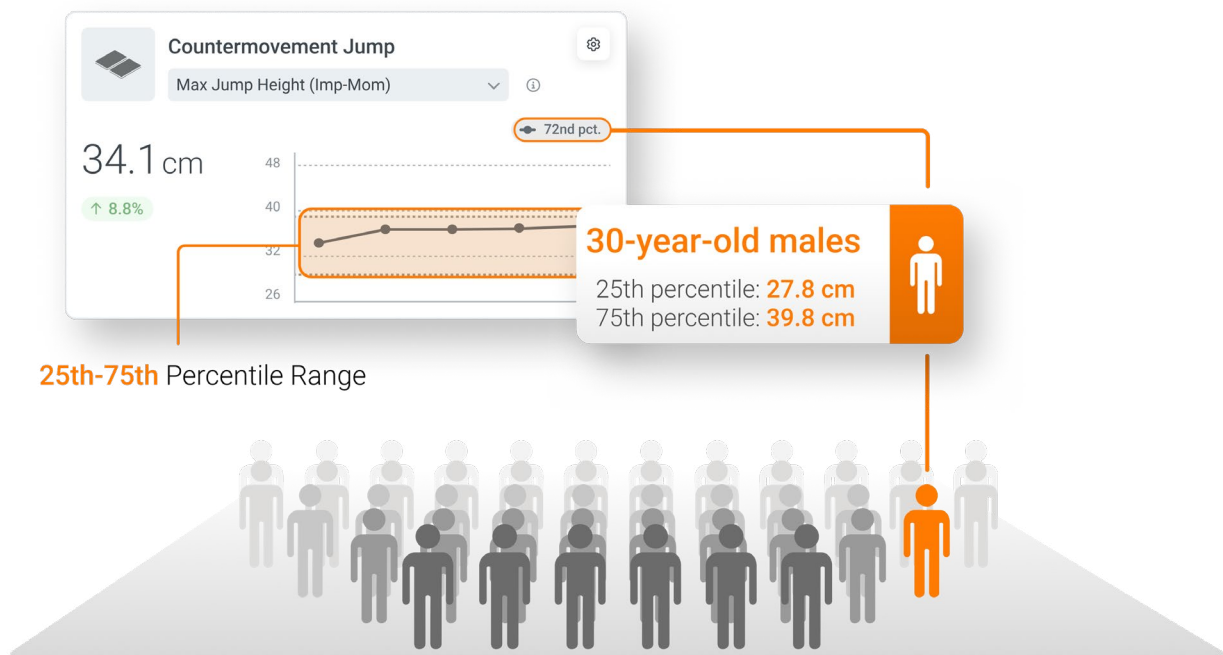
A major opportunity in workplace wellness exists before an injury results in medical escalation, time away from work or a compensation claim. In many workplace settings, physical complaints develop gradually through repeated loading, accumulated fatigue, reduced movement capacity or decreased tolerance to job-specific demands rather than from a single isolated event.



This creates a practical role for objective assessment. By using tools such as force plates, dynamometry systems and motion capture technologies, practitioners can identify reduced capacity, asymmetries or movement limitations before they become more significant barriers to work tolerance.

By using [objective measurement technologies], practitioners can **identify reduced capacity, asymmetries or movement limitations** before they become more significant barriers...

For the practitioner, this means using assessment data to determine who may benefit from further evaluation, which physical qualities warrant intervention and where retesting or program development should be prioritized across individuals, teams or job roles.



For the practitioner... **data [can] determine who may benefit from further evaluation,** which physical qualities warrant intervention and [what]... should be prioritized...

These assessment processes can also support broader health promotion within organizations. Benchmarking against reference values, such as **VALD Norms**, tracking change over time and using simple internal comparisons or **Leaderboards** can help increase engagement and buy-in, particularly when paired with group challenges. In this context, objective testing can support both workforce screening and employee participation.

Internal competitions, such as improving average strength across departments, can further drive participation and reinforce consistent involvement across the workforce. Over time, this creates a structured pathway for ongoing engagement, linking assessment, intervention and retesting to a consistent workforce health strategy.

Workplace Wellness **Service Models**

Practitioners typically deliver workplace wellness services through three primary models. Each represents a different way to engage with organizations, ranging from single-event assessments to fully integrated, ongoing partnerships. These models are not mutually exclusive and often evolve as organizations gain a clearer understanding of workforce capacity and the value of objective testing technologies.

On-site **Testing Events**

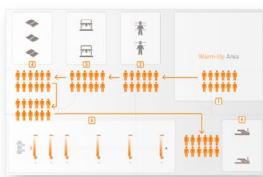
On-site testing events involve delivering group-based assessments directly within the workplace. Employees rotate through a series of structured testing stations, allowing practitioners to assess large groups efficiently within a defined time window. This model prioritizes efficiency and consistency, generating immediate insights at both the individual and organizational levels.

[On-site testing] prioritizes efficiency and consistency, generating immediate insights at both the individual and organizational levels.

Technologies like [HumanTrak](#), [ForceDecks](#) and [SmartSpeed](#) all have group testing capabilities, allowing practitioners to test large groups more quickly and efficiently in on-site settings.

For practitioners looking to implement this model in more detail, the [Practitioner's Guide to Group Testing](#) outlines the key considerations for setup, workflow and scaling.

Free eBook



Practitioner's Guide to **Group Testing**

The definitive guide for practitioners implementing group testing into practice.



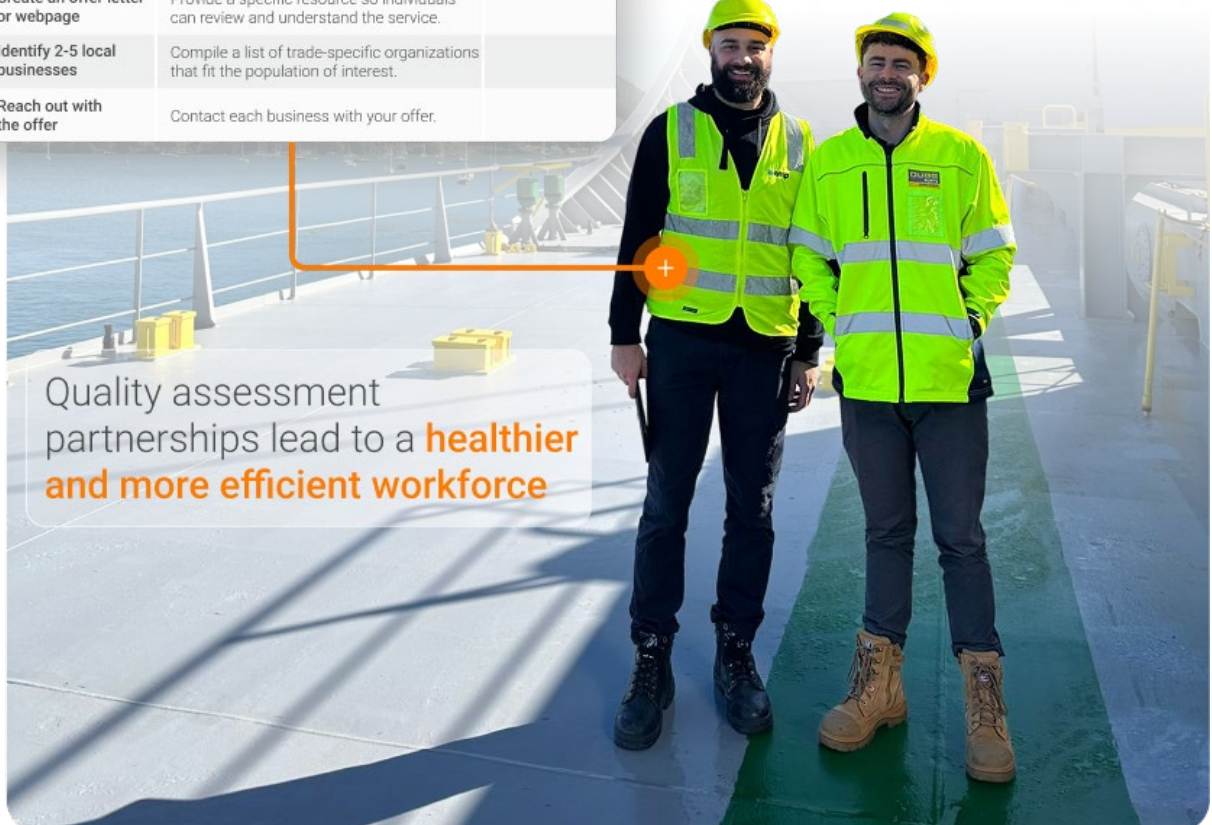
From an organizational perspective, on-site testing events extend beyond data collection by creating employee engagement through testing and reporting strategies. Employees can see how their performance stacks up to reference values such as [VALD Norms](#), while tools like [Leaderboards](#) and platforms like [MoveHealth](#) support consistent comparisons and individualized exercise recommendations.

This approach provides employers with immediate visibility into workforce capacity, establishes a clear pathway for continued engagement and gives practitioners a scalable entry point to demonstrate value.

Workforce Wellness Partnerships

Ongoing wellness partnerships extend beyond single testing events to a more integrated and continuous service model. Within these programs, testing sessions are often delivered in group formats, ranging from large-scale assessments across entire organizations to smaller, team-based sessions. This structure supports efficient data collection during both initial evaluations and re-evaluations while maintaining consistent testing conditions.

Begin Outreach			
✓	Activity	Considerations	Owner
☐	Create an offer letter or webpage	Provide a specific resource so individuals can review and understand the service.	
☐	Identify 2-5 local businesses	Compile a list of trade-specific organizations that fit the population of interest.	
☐	Reach out with the offer	Contact each business with your offer.	



These partnerships typically combine periodic group retesting with the implementation of strength and conditioning programs, ergonomic strategies and injury prevention education. Services like these are often best integrated within a one-stop-shop platform, such as [MoveHealth](#), where practitioners can assign exercise programs, questionnaires and education to help build an individual's overall profile.

...[platforms] such as MoveHealth [allow practitioners to] assign exercise programs, questionnaires and education to **help build an individual's overall profile.**

This model creates consistency for both the practitioner and the organization. Employers benefit from a structured, proactive approach to workforce health, while practitioners can build deeper relationships, demonstrate long-term value and establish recurring revenue streams.

In-Clinic **Services**

Clinic-based models involve employees attending assessments at the practitioner's facility rather than being tested on-site. This approach is often used when a more detailed evaluation is required or when organizations prefer a more individualized service structure. It is also used for specific fitness assessments, such as pre-employment screening, in industries like commercial driving.

Access to a controlled environment and a broader range of equipment allows for more comprehensive assessment when needed. It also provides flexibility in scheduling, particularly for smaller organizations or those without the space to host on-site testing.

Access to a controlled environment and a broader range of equipment **allows for more comprehensive assessment** when needed.



Clinic-based services are commonly used to complement other models, particularly after initial testing events, when specific employees may benefit from further assessment or targeted intervention.

Long-Term Value in Workplace Wellness Programs

Workplace wellness programs are rarely limited to a single testing day. When structured effectively, they often expand into broader service models that support both employee health and organizational decision-making over time.

Expanded Service Options



Exercise Programming

Identify and address movement dysfunction, asymmetries and ROM limitations detected from workplace wellness screens with tailored exercise plans.



Work Conditioning Programs

Implement conditioning programs for specific roles or settings, delivered as required or subsidized services for organizational employees to reduce injury risk.



Repeat Testing Cycles

Schedule recurring testing cycles to ensure beneficial change is occurring and detect potentially harmful changes to prevent future injury.



Pre-Employment Screening

Schedule pre-employment assessments in-clinic for practitioners to assess readiness and reduce the risk of new workplace injury.

Common extensions include repeat testing cycles, targeted exercise programming and work conditioning programs. In some settings, these services may also support pre-employment screening, return-to-work pathways or fitness-for-duty assessment. This creates a more complete model in which testing is not treated as an endpoint, but as the starting point for targeted intervention, retesting and long-term service delivery.

Sustained engagement, however, is driven by demonstrated value rather than service expansion alone. Programs that show measurable improvements in workforce capacity, reductions in injury-related costs or simpler decision-making for employers are more likely to be retained over time.

Sustained engagement, however, is driven by **demonstrated value rather than service expansion alone.**

For practitioners, this means prioritizing clear, customizable reporting in platforms like [VALD Hub](#) and MoveHealth, tracking change and aligning services with outcomes that are meaningful to the organization, rather than simply increasing the volume of services delivered.

Group Testing and Reporting

Test batteries allow practitioners to standardize workplace assessments, improve testing efficiency and ensure the same key outputs are collected across employees and retesting sessions. In workplace settings, this is particularly valuable when testing is delivered in groups or across multiple departments, as it reduces setup variability and improves the consistency of interpretation over time.

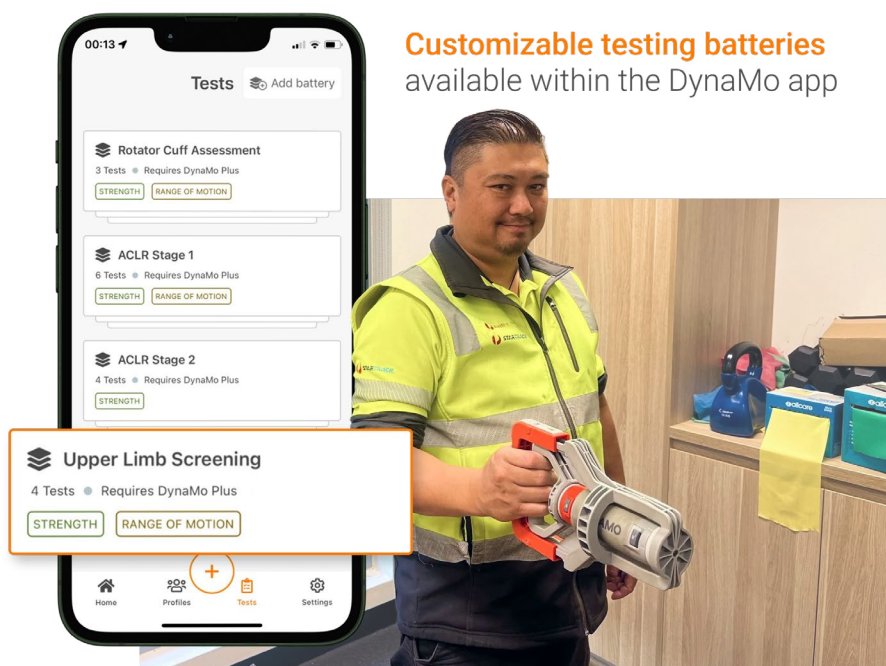
...[test batteries are] particularly valuable...as [they **reduce**] **setup variability and improve the consistency of interpretation** over time.

Test Selection and Group Testing

Workplace test selection should prioritize efficiency, relevance and interpretability. The most effective batteries capture broad indicators of physical capacity while remaining quick to administer across large groups, focusing on assessments that reflect job demands and guide more specific application where needed.

The most effective batteries capture broad indicators of physical capacity while **remaining quick to administer across large groups...**

Group testing workflows support this approach. Technologies such as ForceDecks and HumanTrak enable standardized, high-efficiency testing with minimal setup changes, while DynaMo supports simple, repeatable strength-based batteries across teams or departments. Standardized test batteries improve efficiency, reduce decision fatigue and support consistent retesting.

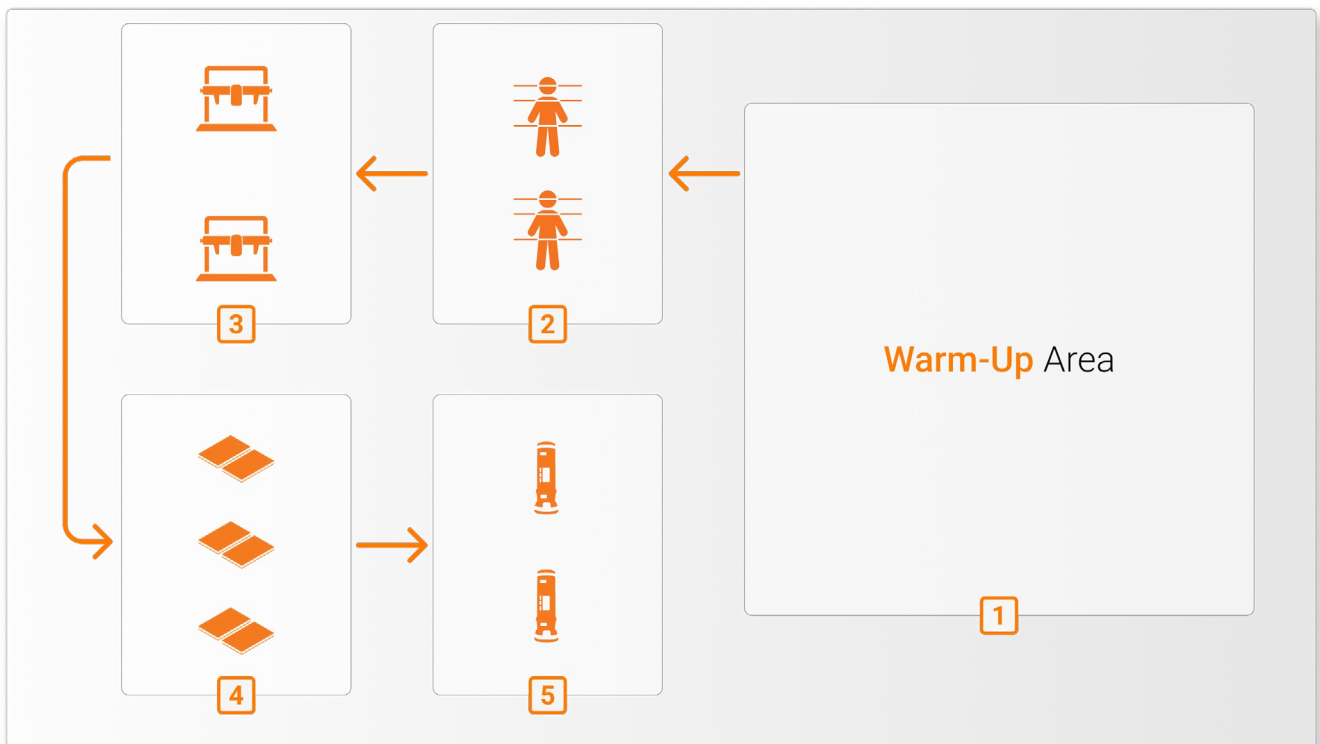


Customizable testing batteries
available within the DynaMo app

A practical workplace battery should focus on broadly relevant, actionable qualities, using the fewest tests necessary to provide meaningful insight into population capacity. Common examples include:

- **Lower-body force capacity** (e.g., isometric mid-thigh pull [IMTP], isometric squat [Iso Squat] or isometric belt squat)
- **Upper-body force capacity** (e.g., grip strength or isometric push-up)
- **Shoulder strength or ROM** (e.g., external-to-internal rotation ratio [ER:IR] or abduction ROM)
- **Movement quality** (e.g., box lift assessment)
- **Landing control or reactive strength** (e.g., land and hold or hop test)

Although some technologies, such as DynaMo, include built-in [test battery functions](#), multi-system test batteries are best implemented using station-specific assessments to maximize testing flow. For more information on how to coordinate high-volume testing scenarios, refer to the [Practitioner's Guide to Group Testing](#).



Similarly, systems like HumanTrak include task-specific assessments such as the [box lift](#), allowing practitioners to evaluate lifting mechanics through measures including spinal flexion during lowering and lifting, trunk extension during overhead pushing and peak hip and knee flexion throughout the task.

Using **Norms** to Support Interpretation

Normative data gives test results context. It allows practitioners to compare performance to expected values and more clearly identify reduced capacity, asymmetries or outlying results that may warrant follow-up.

...[Norms help practitioners] **identify reduced capacity, asymmetries or outlying results** that may warrant follow-up.

Norms can be applied in several ways depending on the workplace setting. Practitioners may compare an employee's results to the following:

- Age- and sex-matched general population reference values
- Internal company or department averages
- Role-specific or task-specific clearance benchmarks

Certain platforms, such as [VALD Hub](#), house their own normative datasets to support comparison with age- and sex-matched data at a global scale. For example, [VALD Norms](#) present large-scale, cleaned datasets as percentiles directly alongside test results, allowing immediate comparison of nearly any test metric available.

Similarly, Normative Data Reports, such as the [2024/25 NFL and NCAA Football Data Report](#), provide broader context on typical performance ranges and commonly used tests in specific performance environments, which can help inform both test selection and interpretation across settings.



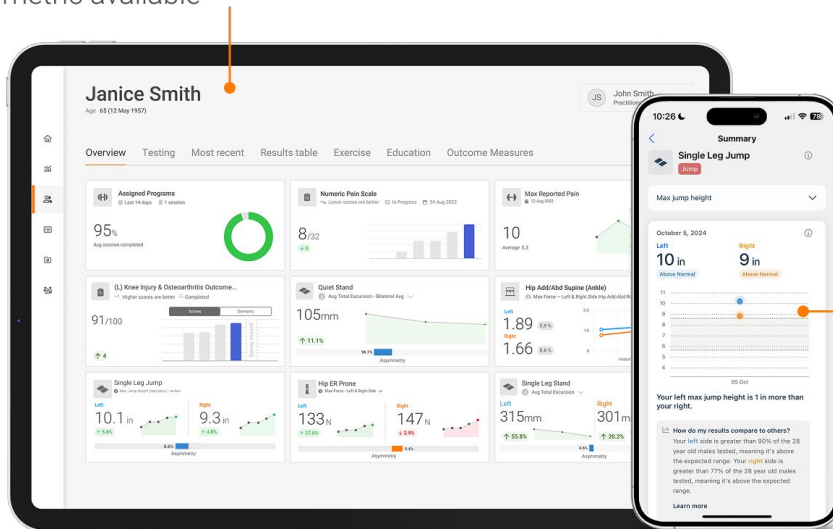
Normative data allows practitioners to identify where an employee may be underperforming relative to peers or expected physical standards. In workplace settings, directly defining “injury risk” is difficult due to the many factors that influence injury occurrence. Using data like VALD Norms provides a clearer basis for intervention and retesting, supporting a growth-oriented model rather than one focused only on screening and reporting.

Extending Value Beyond the Testing Day

A common limitation of workplace testing is that results are often not put into action. [Digital health](#) and exercise prescription platforms like [VALD Hub](#) and MoveHealth can be used to give employees access to their results, monitor progress over time and complete prescribed exercises outside the testing session.

Practitioner View

Detailed, longitudinal data and comparative Norms for any metric available



Participant View

Simplified metric lists and normative data explanations for better understanding of performance results

...VALD Hub and MoveHealth can be used to give employees access to their results, **monitor progress over time and complete prescribed exercises...**

This helps practitioners extend workplace wellness beyond the initial screen and provides a simple way to support follow-up, retesting and ongoing exercise delivery. It also allows workplace programs to function as more than a single testing event by creating continuity between assessment and intervention.

Testing by Population

Test selection should be adapted to the specific physical demands of each workplace population to ensure assessments remain relevant, efficient and actionable.

Warehouse and Manual Handling Workers

Warehouse and manual handling environments often involve repetitive lifting, carrying, reaching and transitioning in and out of varied working positions. A practical battery may therefore prioritize:

Test	System	Metric
IMTP	ForceDecks	Peak vertical force and asymmetry
Box Lift Assessment	HumanTrak	Trunk flexion at peak knee flexion during lift
Shoulder Abduction ROM	DynaMo	Total ROM and asymmetry to identify potential limitations for overhead lifting tasks
Grip Test	DynaMo	Peak force

This type of battery helps practitioners assess whether employees demonstrate the baseline physical qualities needed to tolerate repetitive loading and transitional movement demands.



Quadrant plots help **organize high and low performers** across different tests

In Dashboard visualizations, [quadrant plots](#) can be used to compare metrics across groups. This helps identify individuals or teams who may benefit from targeted intervention.

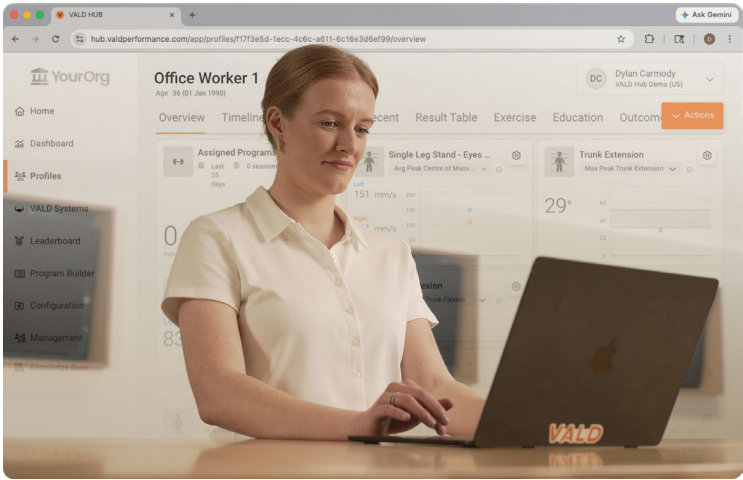
Corporate Office Workers

Corporate office populations typically present with lower absolute physical demand but often demonstrate reduced movement variability, lower strength capacity and limitations in mobility or postural control. A practical battery may include:

Test	System	Metric
Squat Assessment	ForceDecks	Eccentric and concentric strength and asymmetry, as well as squat depth
Hand Grip and Shoulder External Rotation	DynaMo	Peak force
Quiet Stand and Single Leg Stand (Balance)	HumanTrak	Center of pressure ellipse
Box Lift	HumanTrak	Trunk flexion at peak knee flexion during lift

In this setting, the goal is often less about high-load occupational performance and more about identifying general deconditioning, mobility restrictions and asymmetries that may contribute to discomfort or reduced physical capacity over time.

In this setting, the goal is...about **identifying general deconditioning, mobility restrictions and asymmetries** that may contribute to...reduced physical capacity over time.

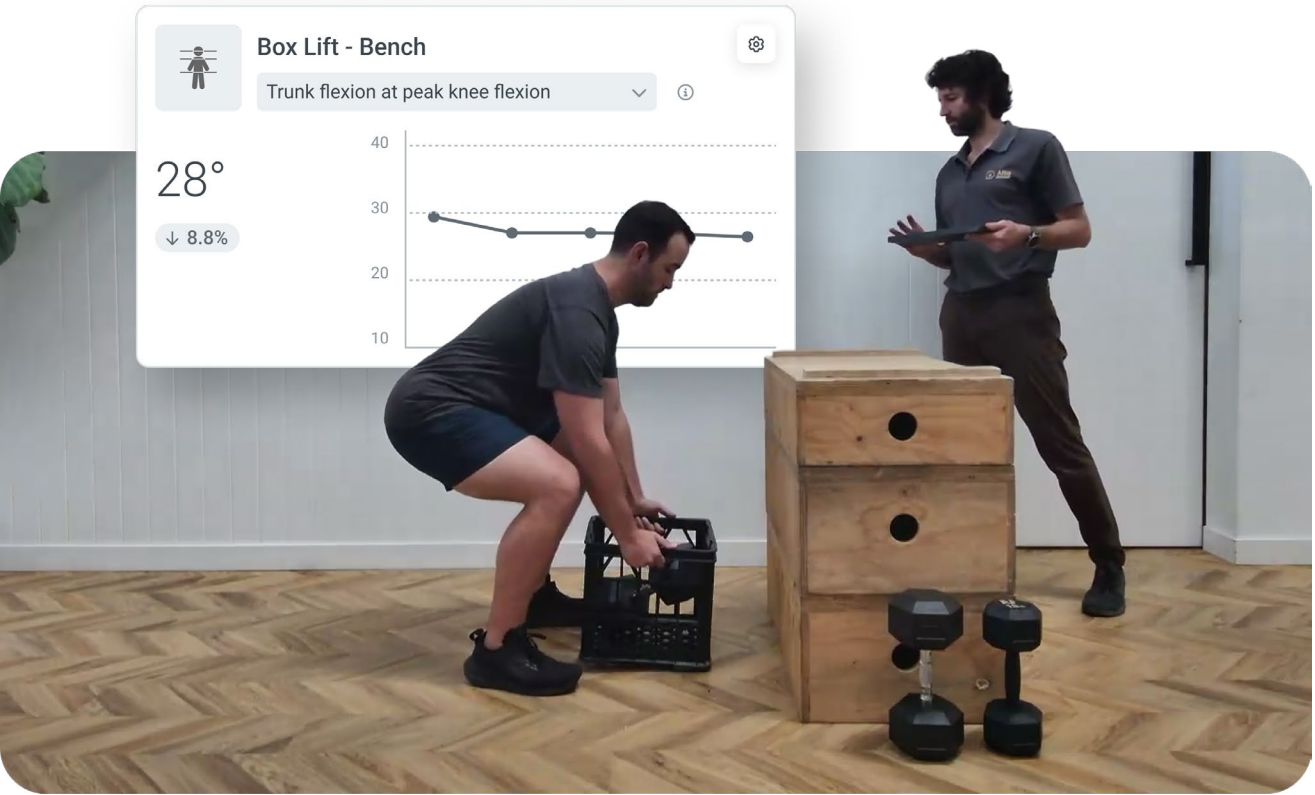


Efficient workplace wellness programs are built on simple test batteries, clear interpretation and actionable follow-up. When test selection, normative context and digital follow-up are combined effectively, practitioners can deliver screening models that are both scalable and meaningful across a wide range of workplace populations.

Construction Workers

Construction environments often involve high-force, multi-planar movements, including lifting, carrying, climbing, working overhead for sustained periods and operating in unstable or uneven conditions. A practical battery may therefore prioritize:

Test	System	Metric
IMTP	ForceDecks	Peak vertical force and asymmetry
Box Lift Assessment	HumanTrak	Trunk flexion at peak knee flexion during lift
Shoulder Strength or ROM	HumanTrak	Internal rotation (IR), external rotation (ER) and abduction to assess overhead capacity
Single Leg Balance or Land and Hold	HumanTrak	95% center of pressure ellipse and control in variable environments
Grip Test	DynaMo	Peak force for tool handling and sustained load tolerance



This type of battery helps practitioners assess whether employees demonstrate the strength, control and stability required to tolerate the high-load and variable task demands common in construction settings.

Case Study

EMP Onsite: Ports and Shipping Docks

[EMP Onsite](#) implemented an injury prevention program at [QUBE Ports](#) using objective strength data collected through [VALD technology](#) to improve how musculoskeletal injury risk was managed. By applying a preventive model aligned with strength and conditioning, functional assessments were used to quantify how employees move and produce force. This allowed practitioners to identify specific physical deficits and develop targeted conditioning programs to improve capacity before injury occurred.



In ports and shipping environments, cargo is often lifted out of the vessel using hoists, cranes and other rigging gear. This requires teams to climb, stand or even walk on objects such as steel, timber and other hard goods, depending on where and how lifting points need to be accessed. Balance is an essential physical trait to assess, given the risk of slips, trips or falls in this work scenario.

Shoulder function was a primary focus, with 26% of employees reporting symptoms likely linked to repetitive, high-load tasks such as lashing and unlashings. Testing consistently identified reduced ER and IR strength across sites, leading to targeted exercise and on-site conditioning interventions.

...26% of employees [reported shoulder] symptoms likely linked to repetitive, high-load tasks such as lashing and unlashings.

While shoulder strength remains a key priority, assessments such as IMTP provide additional insight into overall physical capacity, with higher peak force associated with lower perceived effort, reduced fatigue and improved resilience to injury in physically demanding port environments.

Test	Metric	Rationale
Shoulder ER and IR	Shoulder ER peak force	Shoulder injuries make up a significant percentage of workplace injuries
	Shoulder IR peak force	
	Shoulder ER:IR ratio	
Pinch Grip	Peak force	Overall indicator of upper-body strength and longevity
IMTP	Peak vertical force	Indicator of lower-body strength, asymmetry and relative effort during work tasks
	Peak vertical force asymmetry	

This resulted in measurable improvements in physical capacity, with IR strength increasing by 11% (17.2kg to 19.4kg) and ER by 19.8% (13.1kg to 15.7kg). Normative comparisons were then used to stratify risk and guide ongoing programming.

Importantly, strength alone does not reflect shoulder health. The balance between ER and IR strength helps inform injury risk, with an IR:ER ratio of 1.3 linked to reduced injury risk in overhead tasks ([Asker et al., 2018](#); EMP Onsite data). When both values are within ~10% of normative ranges, the shoulder is better positioned to avoid impingement and poor mechanics. This is particularly important in lifting-heavy environments, where shoulder injuries are common.

These improvements were associated with an estimated reduction of four shoulder-related claims, equivalent to approximately \$144,000 in direct cost savings. More broadly, the program contributed to an estimated annual return on investment (ROI) of 5.5:1, with further projected reductions in injury frequency and workers' compensation costs approaching \$900,000.

More broadly, the program contributed to an **estimated annual ROI of 5.5:1**...[reducing] workers' compensation costs [by almost] \$900,000.

What Next?

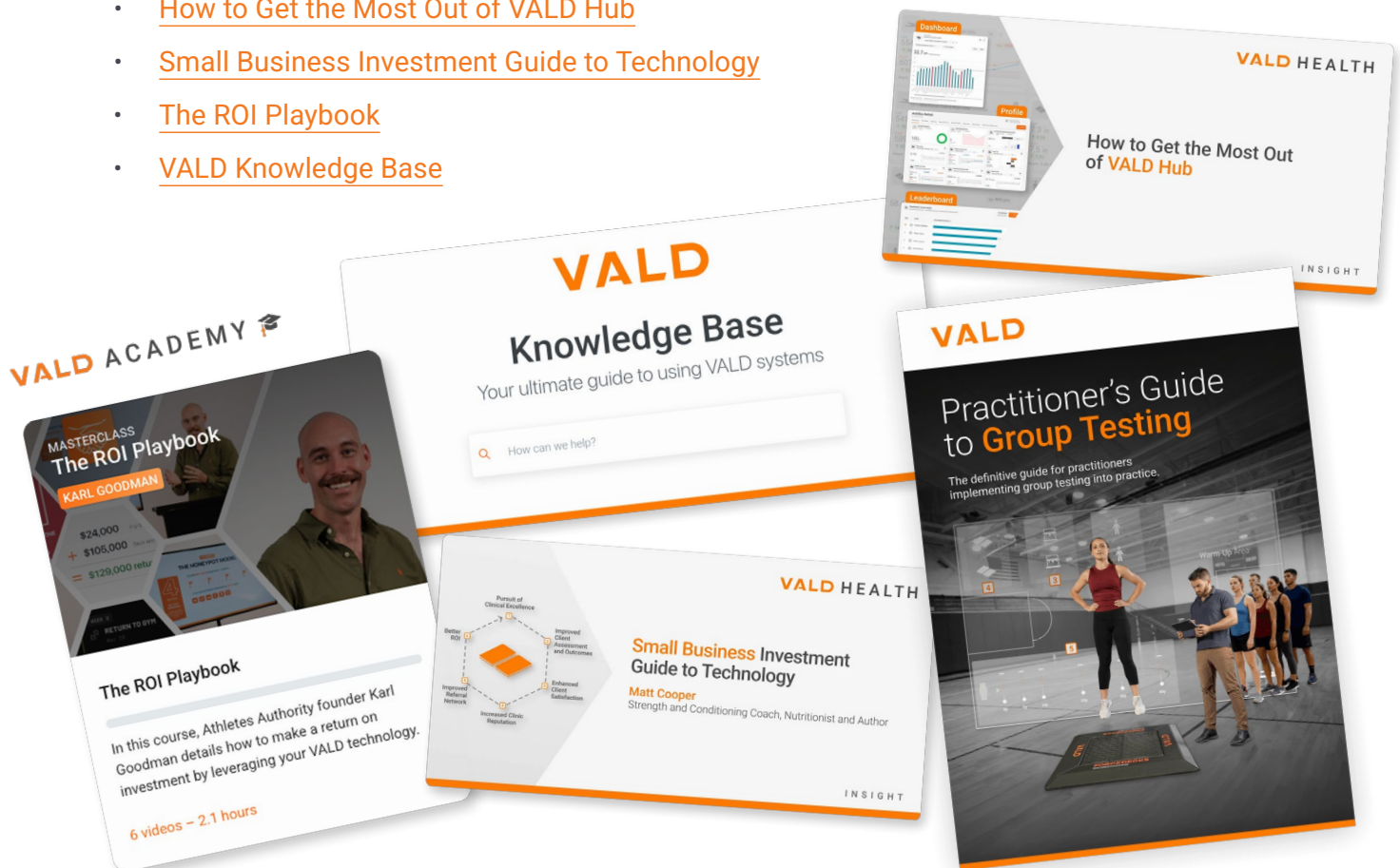
Workplace wellness programs can become a meaningful and scalable service line for practitioners, particularly when they move beyond single testing events into structured, repeatable systems. The long-term value lies in building processes that support ongoing assessment, intervention and retesting, allowing practitioners to consistently engage with organizations and demonstrate change over time.

For practitioners looking to begin, the next step is to identify organizations whose workforce demands align with their service model and implement a simple, efficient testing process. From this starting point, programs can expand through repeat testing cycles, targeted training, workplace-specific exercise initiatives and education.

When supported by clear reporting and practical application, these services create a foundation for long-term partnerships by helping organizations reduce injury-related costs, guide targeted intervention and track change over time, positioning workplace wellness as a sustainable extension of clinical or performance practice.

To support these next steps, the following resources may be helpful:

- [Practitioner's Guide to Group Testing](#)
- [Practitioner's Guide to ROI](#)
- [How to Get the Most Out of VALD Hub](#)
- [Small Business Investment Guide to Technology](#)
- [The ROI Playbook](#)
- [VALD Knowledge Base](#)



Appendix

Best Practice Checklist

Downloadable version

Define Your Service Offering

✓	Activity	Considerations	Owner
☐	Identify the type of service offered	Onsite, recurring wellness program or clinic-based employee assessments.	
☐	Determine the population of interest	Assess unique regional trades and identify an at-risk population.	
☐	Identify timing of delivery	Decide on frequency of assessments (if using a recurring model).	
☐	Determine personnel requirements	Consider whether multiple personnel may be required to provide your service.	

Build a Practical Assessment Battery

✓	Activity	Considerations	Owner
☐	Balance	Determine an activity-appropriate balance assessment to help identify falls risk.	
☐	Strength	Select task-specific strength tests to assess readiness for activity.	
☐	Power	Select task-specific power tests for unique workplace demands, if relevant.	
☐	Functional task	Assess kinematics like lifting technique to potentially help flag injury risk.	

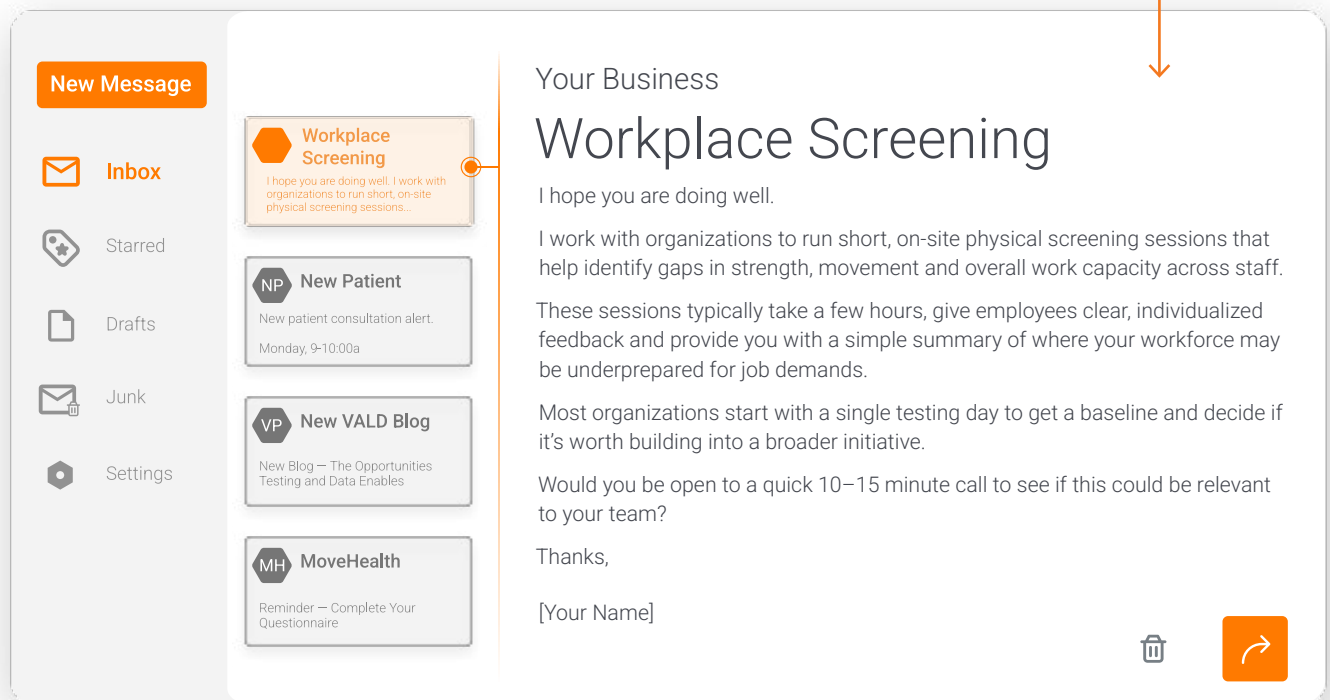
Creating a Reporting Flow

✓	Activity	Considerations	Owner
☐	Select relevant metrics	Determine which metrics will be used to build each group and individual profile.	
☐	Individual summaries	Select a health platform (e.g., MoveHealth) to deliver individual testing information.	
☐	Group summaries	Select a reporting platform (e.g., VALDHub) to create group summaries and reports.	

Begin Outreach

✓	Activity	Considerations	Owner
☐	Create an offer letter or webpage	Provide a specific resource so individuals can review and understand the service.	
☐	Identify 2-5 local businesses	Compile a list of trade-specific organizations that fit your population of interest.	
☐	Reach out with the offer	Contact each business with your offer.	

Outreach Email



Testing Timeline (Warehouse Setting)

 Downloadable version

Time	Group A	Group B	Group C	Group D
08:00-08:15	Group warm-up	Group warm-up	Group warm-up	Group warm-up
08:15-08:30	Station 1: IMTP (ForceDecks)	Station 2: ER strength (DynaMo)	Station 3: Lunge assessment (HumanTrak)	Station 4: Land and hold (ForceDecks)
08:30-08:45	Station 2: ER strength (DynaMo)	Station 3: Lunge assessment (HumanTrak)	Station 4: Land and hold (ForceDecks)	Station 1: IMTP (ForceDecks)
08:45-09:00	Station 3: Lunge assessment (HumanTrak)	Station 4: Land and hold (ForceDecks)	Station 1: IMTP (ForceDecks)	Station 2: ER strength (DynaMo)
09:00-09:15	Station 4: Land and hold (ForceDecks)	Station 1: IMTP (ForceDecks)	Station 2: ER strength (DynaMo)	Station 3: Lunge assessment (HumanTrak)
09:15-09:30	Buffer/exit	Buffer/exit	Buffer/exit	Buffer/exit

Occupational health doesn't need to be **reactive**.

Musculoskeletal injuries remain one of the leading drivers of lost work time and cost across industries. Despite this, most workplace health strategies still rely on reactive care, self-reported symptoms or generalized wellness initiatives.

This guide outlines how practitioners can apply objective assessment to workplace environments, using structured testing, clear reporting and targeted intervention to better understand and develop workforce capacity. By aligning assessment with job demands, practitioners can support injury reduction, improve performance and create scalable service models within occupational settings.

Some questions we'll answer in this guide:

- Q: How can I implement objective testing in workplace settings?
- Q: Which assessments are most relevant for different workforces?
- Q: How do I interpret results using tools such as VALD Norms?
- Q: How can I structure services to create long-term partnerships?
- Q: What does a scalable workplace wellness model look like?



Digital
download

Want to learn more? We are here to help.

info@vald.com